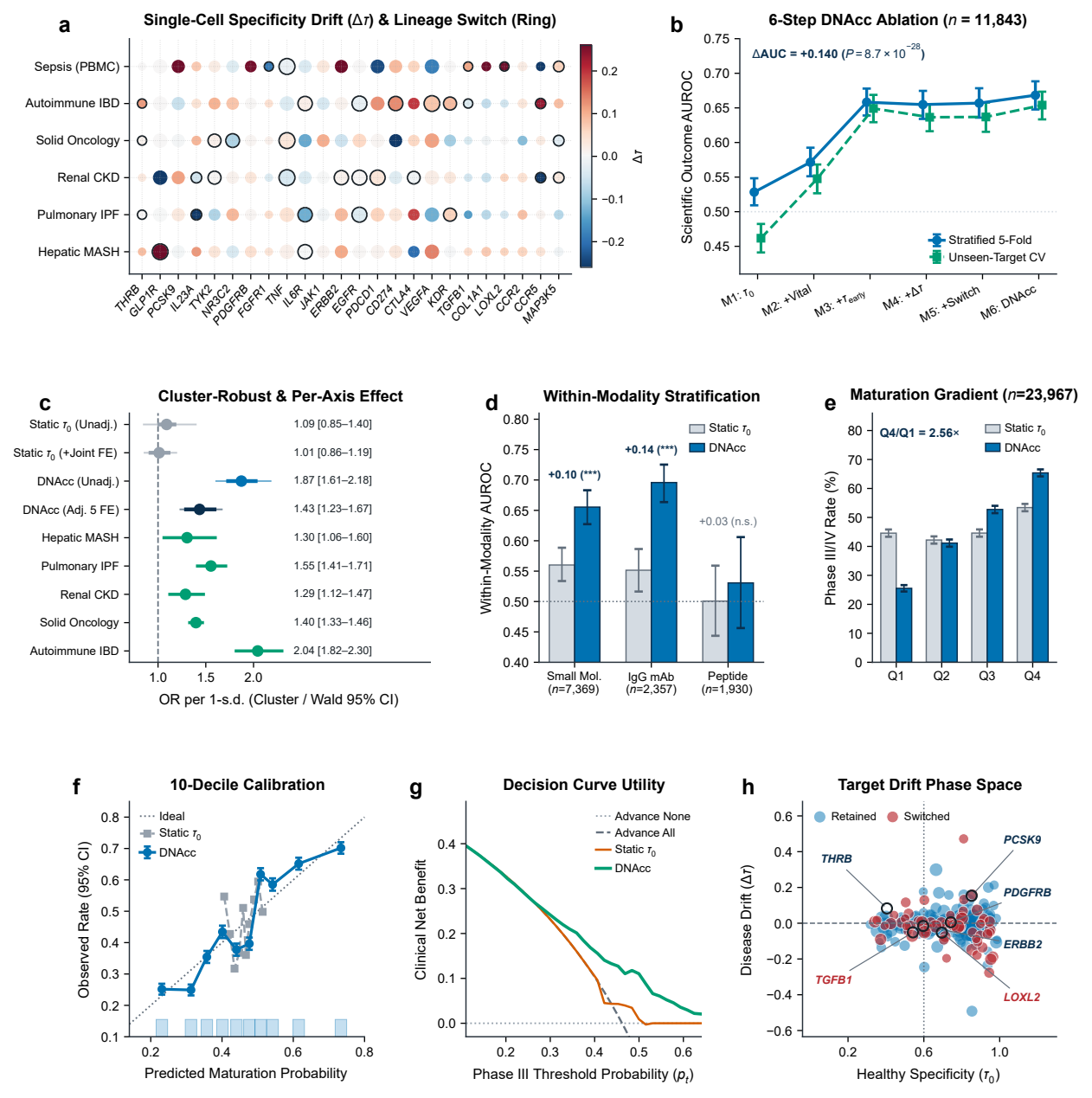

a
Single-Cell Specificity Drift (Δτ) & Lineage Switch (Ring)
Sepsis (PBMC)
Autoimmune IBD
Solid Oncology
Renal CKD
Pulmonary IPF
Hepatic MASH
THRB GLP1R PCSK9 IL23A TYK2 NR3C2 PDGFRB FGFR1 TNF IL6R JAK1 ERBB2 EGFR PDCD1 CD274 CTLA4 VEGFA KDR TGFB1 COL1A1 LOXL2 CCR2 CCR5 MAP3K5
0.2
0.1
0.0
−0.1
−0.2
Δτ
b
6-Step DNAcc Ablation (n = 11,843)
ΔAUC = +0.140 (P = 8.7 × 10−28)
Scientific Outcome AUROC
0.75
0.70
0.65
0.60
0.55
0.50
0.45
Stratified 5-Fold
Unseen-Target CV
M1: τ0
M2: +Vital
M3: +Tearly
M4: +Δτ
M5: +Switch
M6: DNAcc
c
Cluster-Robust & Per-Axis Effect
Static τ0 (Unadj.)
Static τ0 (+Joint FE)
DNAcc (Unadj.)
DNAcc (Adj. 5 FE)
Hepatic MASH
Pulmonary IPF
Renal CKD
Solid Oncology
Autoimmune IBD
1.09 [0.85–1.40]
1.01 [0.86–1.19]
1.87 [1.61–2.18]
1.43 [1.23–1.67]
1.30 [1.06–1.60]
1.55 [1.41–1.71]
1.29 [1.12–1.47]
1.40 [1.33–1.46]
2.04 [1.82–2.30]
1.0
1.5
2.0
OR per 1-s.d. (Cluster / Wald 95% CI)
d
Within-Modality Stratification
Within-Modality AUROC
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
0.40
Static τ0
DNAcc
+0.10 (***)
+0.14 (***)
+0.03 (n.s.)
Small Mol. (n=7,369)
IgG mAb (n=2,357)
Peptide (n=1,930)
e
Maturation Gradient (n=23,967)
Q4/Q1 = 2.56×
Static τ0
DNAcc
Phase III/IV Rate (%)
80
70
60
50
40
30
20
10
0
Q1
Q2
Q3
Q4
f
10-Decile Calibration
Ideal
Static τ0
DNAcc
Observed Rate (95% CI)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.2
0.4
0.6
0.8
Predicted Maturation Probability
g
Decision Curve Utility
Advance None
Advance All
Static τ0
DNAcc
Clinical Net Benefit
0.4
0.3
0.2
0.1
0.0
0.2
0.3
0.4
0.5
0.6
Phase III Threshold Probability (pt)
h
Target Drift Phase Space
Retained
Switched
PCSK9
THRB
PDGFRB
ERBB2
TGFB1
LOXL2
Disease Drift (Δτ)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.2
0.6
1.0
Healthy Specificity (τ0)